

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q18-0970

Aguas Buenas Gis Tc 138 kV  
April 26, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0970. The Interconnection Customer proposes a 346.4 MW solar photovoltaic facility interconnecting at the Aguas Buenas Gis Tc 138 kV substation in the Tennessee area. The request was received on March 28, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 401 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$79,239,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                    | Contingency                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------------|-----------------------------------------------|-------------|--------------|----------|
| COYOTE to HOBART 138 kV                               | CLINCH RIV to CARLTON HILLS 345 kV            | 150.7       | 253          | 5.8      |
| ROCHESTER to RADFORD ARMY AMMUNIION PLANT 138 kV Line | LOWMOOR to HINTON AP 345 kV                   | 145.1       | 183          | 5.5      |
| SYSQUOC to ROCHESTER 138 kV Line                      | LOWMOOR to HINTON AP 345 kV                   | 134.8       | 209          | 5.5      |
| TALOGA to POMONA HEIGHTS 138 kV                       | HINTON AP to LOWMOOR 345 kV                   | 131.5       | 252          | 5.5      |
| SVP to INGLES ST SUBSTATION 138 kV                    | LOWMOOR to HINTON AP 345 kV line              | 130.1       | 191          | 5.5      |
| N CLAYTOR to SVP 138 kV Line                          | LOWMOOR to HINTON AP SUBSTATION 345 kV line   | 128.8       | 193          | 5.5      |
| LEBANON to WINNEBAGO WIND POWER PROJECT 138 kV        | BARR LAKE to KEAN MTN AND GRASSY CRK 345 kV   | 127.1       | 242          | 6.6      |
| N CLAYTOR to RADFORD ARMY AMMUNITION PLANT 138 kV     | LOWMOOR to HINTON AP 345 kV                   | 127.0       | 211          | 5.5      |
| BYHALIA to LEBANON SUBSTATION 138 kV                  | KEAN MTN AND GRASSY CRK to BARR LAKE 345 kV L | 125.7       | 235          | 6.6      |
| MCCLAIN ENERGY FACILITY to COAL 138 kV                | HINTON AP to MANATEE 345 kV line              | 125.4       | 374          | 6.0      |
| S. LAND PARK & 35TH AVE to PRINGLE 138 kV             | NEW HOPE AP to MANATEE 345 kV                 | 125.3       | 216          | 6.8      |
| CALPEAK PE 7 SUBSTATION to BYHALIA 138 kV             | BARR LAKE to KEAN MTN AND GRASSY CRK 345 kV   | 124.0       | 231          | 6.6      |
| HINTON AP 345/138kV Transformer                       | MANATEE to HINTON AP 345 kV                   | 122.9       | 640          | 11.4     |

| Monitored Facility                                 | Contingency                                           | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------|-------------------------------------------------------|-------------|--------------|----------|
| SYSQUOC to LAKE FORREST 138 kV                     | LOWMOOR to HINTON AP 345 kV                           | 121.9       | 243          | 5.5      |
| LAKE FORREST to LYTLE 138 kV                       | LOWMOOR to HINTON AP 345 kV Line                      | 120.1       | 256          | 5.5      |
| POMONA HTS to LYTLE SUBSTATION 138 kV line         | LOWMOOR to HINTON AP 345 kV line                      | 118.5       | 270          | 5.5      |
| SCANLON to AGUAS BUENAS GIS TC SUBSTATION 138 kV   | SANTA ANA 3 to AGUAS BUENAS GIS TC 138 kV             | 117.4       | 226          | 79.0     |
| WINNEBAGO WIND POWER PROJECT to CLINCH RIV 138 kV  | KEAN MTN AND GRASSY CRK to BARR LAKE 345 kV           | 116.7       | 273          | 6.6      |
| ASPEN to LOWMOOR 345 kV                            | BATH COUNTY to LONGVIE2 345 kV line                   | 116.1       | 1,321        | 17.4     |
| HUBBARDSTOWN to TRISTATE 345 kV                    | INTERNATIONAL PAPER to HANGING ROCK SUBSTATION 500 kV | 114.5       | 1,167        | 6.1      |
| RED HILLS SUBSTATION to VASEAUX LAKE 138 kV L      | LOWMOOR to HINTON AP 345 kV                           | 113.3       | 519          | 8.0      |
| MCCAIN ENERGY FACILITY to VASEAUX LAKE 138 kV Line | LOWMOOR to HINTON AP 345 kV Line                      | 113.2       | 522          | 8.0      |
| HALLS RIDGE SUBSTATION to PRINGLE 138 kV           | NEW HOPE AP SUBSTATION to MANATEE 345 kV              | 112.1       | 243          | 6.8      |
| LOWMOOR to HINTON AP 345 kV                        | MCCLAIN ENERGY FACILITY to VASEAUX LAKE 138 kV        | 111.7       | 1,182        | 21.5     |
| PALERMO to SCANON 138 kV                           | SANTA ANA 3 to AGUAS BUENAS GIS TC 138 kV             | 110.7       | 230          | 79.0     |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| COYOTE to HOBART 138 kV                           | Rebuild line with higher-capacity conductor | \$5,484,000   | 23% (\$1,246,000)     |
| ROCHESTER to RADFORD ARMY AMMUNITION PLANT 138 kV | Rebuild line with higher-capacity conductor | \$9,497,000   | 23% (\$2,176,000)     |
| ROCHESTER to SYSQUOC 138 kV                       | Rebuild line with higher-capacity conductor | \$5,797,000   | 23% (\$1,328,000)     |
| POMONA HTS to TALOGA 138 kV line                  | Rebuild line with higher-capacity conductor | \$30,245,000  | 23% (\$6,929,000)     |
| SVP to INGLES ST 138 kV                           | Rebuild line with higher-capacity conductor | \$1,739,000   | 23% (\$398,000)       |
| N CLAYTOR to SVP 138 kV Line                      | Rebuild line with higher-capacity conductor | \$4,486,000   | 23% (\$1,028,000)     |
| LEBANON to WINNEBAGO WIND POWER PROJECT 138 kV    | Rebuild line with higher-capacity conductor | \$2,758,000   | 33% (\$921,000)       |

| Facility                                                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| NORTH CLAYTOR to RADFORD ARMY AMMUNITION PLANT 138 kV line | Rebuild line with higher-capacity conductor | \$3,178,000   | 23% (\$728,000)       |
| BYHALIA to LEBANON SUBSTATION 138 kV                       | Rebuild line with higher-capacity conductor | \$10,629,000  | 33% (\$3,551,000)     |
| COAL to MCCLAIN ENERGY FACILITY 138 kV                     | Rebuild line with higher-capacity conductor | \$7,713,000   | 23% (\$1,745,000)     |
| S. LAND PARK & 35TH AVE to PRINGLE 138 kV                  | Rebuild line with higher-capacity conductor | \$4,217,000   | 20% (\$850,000)       |
| CALPEAK PE 7 SUBSTATION to BYHALIA 138 kV                  | Rebuild line with higher-capacity conductor | \$16,887,000  | 33% (\$5,641,000)     |
| HINTON AP SUBSTATION 345-138 kV Transformer Bank           | Replace transformer with higher-rated unit  | \$21,773,000  | 13% (\$2,763,000)     |
| SYSQUOC to LAKE FORREST 138 kV                             | Rebuild line with higher-capacity conductor | \$2,483,000   | 23% (\$569,000)       |
| LAKE FORREST to LYTLE 138 kV line                          | Rebuild line with higher-capacity conductor | \$3,866,000   | 23% (\$886,000)       |
| POMONA HTS to LYTLE SUBSTATION 138 kV line                 | Rebuild line with higher-capacity conductor | \$3,597,000   | 23% (\$824,000)       |
| AGUAS BUENAS GIS TC to SCANLON 138 kV Line                 | Rebuild line with higher-capacity conductor | \$5,108,000   | 94% (\$4,823,000)     |
| WINNEBAGO WIND POWER PROJECT to CLINCH RIV 138 kV          | Rebuild line with higher-capacity conductor | \$6,130,000   | 33% (\$2,048,000)     |
| ASPEN to LOWMOOR 345 kV                                    | Rebuild line with higher-capacity conductor | \$83,602,000  | 13% (\$10,771,000)    |
| HUBBARDSTOWN to TRISTATE 345 kV                            | Rebuild line with higher-capacity conductor | \$46,167,000  | 6% (\$2,748,000)      |
| RED HILLS SUBSTATION to VASEAUX LAKE 138 kV L              | Rebuild line with higher-capacity conductor | \$13,102,000  | 19% (\$2,439,000)     |
| VASEAUX LAKE to MCCLAIN ENERGY FACILITY 138 kV             | Rebuild line with higher-capacity conductor | \$9,908,000   | 19% (\$1,845,000)     |
| PRINGLE to HALLS RIDGE 138 kV L                            | Rebuild line with higher-capacity conductor | \$6,799,000   | 20% (\$1,371,000)     |
| LOWMOOR to HINTON AP 345 kV                                | Rebuild line with higher-capacity conductor | \$127,443,000 | 12% (\$15,980,000)    |
| PALERMO to SCANON 138 kV                                   | Rebuild line with higher-capacity conductor | \$5,965,000   | 94% (\$5,631,000)     |
|                                                            | Total allocated to project                  |               | \$79,239,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.